
EE 602: Control System Applications

Course Code	EE 602
Course Name	Control System Applications
L-T-P-C	2.5-0.5-0-3
Prerequisites	Consent of the faculty member
Intended for	MS/PhD
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Basic concepts:** Introduction, basic terminology, objective of subject, some basic examples, Notion of feedback; open- and closed-loop systems.
- **Mathematical Models:** Representation of physical systems and analogous systems, Laplace transforms, block diagrams, transfer functions for different type of systems, block diagrams reduction techniques; Signal flow graphs and Masons gain formula.
- **Control hardware and their models:** Potentiometers, synchros, LVDT, DC and AC servo motors, tachogenerators, electro-hydraulic valves, and pneumatic actuators.
- **Time-domain analysis:** Time domain performance criterion, transient response of first order, second order and higher order systems; Steady state errors: Static and dynamic error constants, system types, steady state errors for unity and non unity feedback systems, performance analysis for P, PI and PID controllers.
- **Frequency-domain analysis:** Bode and polar plots, frequency-domain specifications, correlation between transient response and frequency response.
- **Stability analysis:** Concept of stability by Routh stability criterion, Nyquist stability criterion, gain and phase margins, relative stability, constant M and N circles, Nichols chart and its application.
- **Root-locus technique:** Nature of root-locus, rules of construction, root-locus analysis of control systems.
- **Compensation:** Types of compensation, Proportional, PI and PID controllers; Lead-lag compensators.
- **State-space concepts:** Eigen values and eigen vectors; Solution of state equations; Controllability; Observability; pole placement result, Minimal representations.

Text and Reference Books

1. Nagrath I. J. and Gopal M., **Control System Engineering**.
2. Kuo B. C., **Automatic Control Systems**.
3. Ogata K., **Modern Control Engineering**.
4. Gopal M., **Control Systems: Principle and Design**.